



BITS Pilani
Pilani | Dubai | Goa | Hyderabad

Prayogshala Cluster Lab Manual

CSIS Dept, WILP

<https://prayogshala.bits-pilani.ac.in>



Important instructions

-  Save all your work in the /workspace folder before stopping the cluster. Files stored outside /workspace may be permanently lost when the VM is terminated.
-  Keep the lab session tab open. The session requires regular activity/heartbeat and may terminate after inactivity.
-  A cluster consists of a Master node and Worker nodes. Use the available node controls only when required for your lab activity.
-  Launching all nodes may take a few minutes. Please remain on the page while the cluster is being provisioned.
-  Stop all nodes when you have completed your work but need to retain the session and this will be retained for 15mins only. Terminate all nodes only when you have finished the lab and no longer need the environment.
-  Experiencing a lab launch issue? If your lab fails to launch, close the current session and try again after a few minutes.  Occasionally, such issues may occur due to temporary AWS infrastructure availability.

Step 1 - Open Prayogshala Portal

Click 'Continue with BITS SSO'.

The screenshot shows a web browser window with the address bar displaying `https://stg.bitslabs.in/auth/login`. The page features the BITS logo and the text "WORK INTEGRATED LEARNING PROGRAMMES" and "Prayogshala". Below this, a description states: "Hands-on engineering, without the campus. Work directly on high-end, industry-grade infrastructure — GPU clusters, cloud machines and real lab hardware — provisioned for your programme and ready in your browser, wherever you work." There is a section for "First time here?" with the text "Lab access is provisioned by your department coordinator." and a link for "Contact IT". Below this are five lab categories: Virtual Labs (Cloud VMs), Cluster Labs (GPU / CPU Containers), GPU Labs (On-demand GPUs), Remote Labs (Physical Hardware), and Console Labs (Cloud Sandbox). On the right side, there is a "Sign in to BITS Prayogshala" box. Inside this box, there is a blue button labeled "Continue with BITS SSO" with a right arrow. A red arrow points from the text "Click here" to this button. Below the button, it says "You'll be redirected to the BITS Pilani identity provider, then returned here." At the bottom of the page, there are links for "Help Center", "Privacy Policy", and "Contact IT", followed by a disclaimer: "An institution deemed to be a University estd. vide Sec.3 of the UGC Act, 1956." The Windows taskbar at the bottom shows the date as 30-07-2026 and time as 11:47.

Sign in
to **BITS Prayogshala**

Secured by BITS Single Sign-On

Continue with BITS SSO →

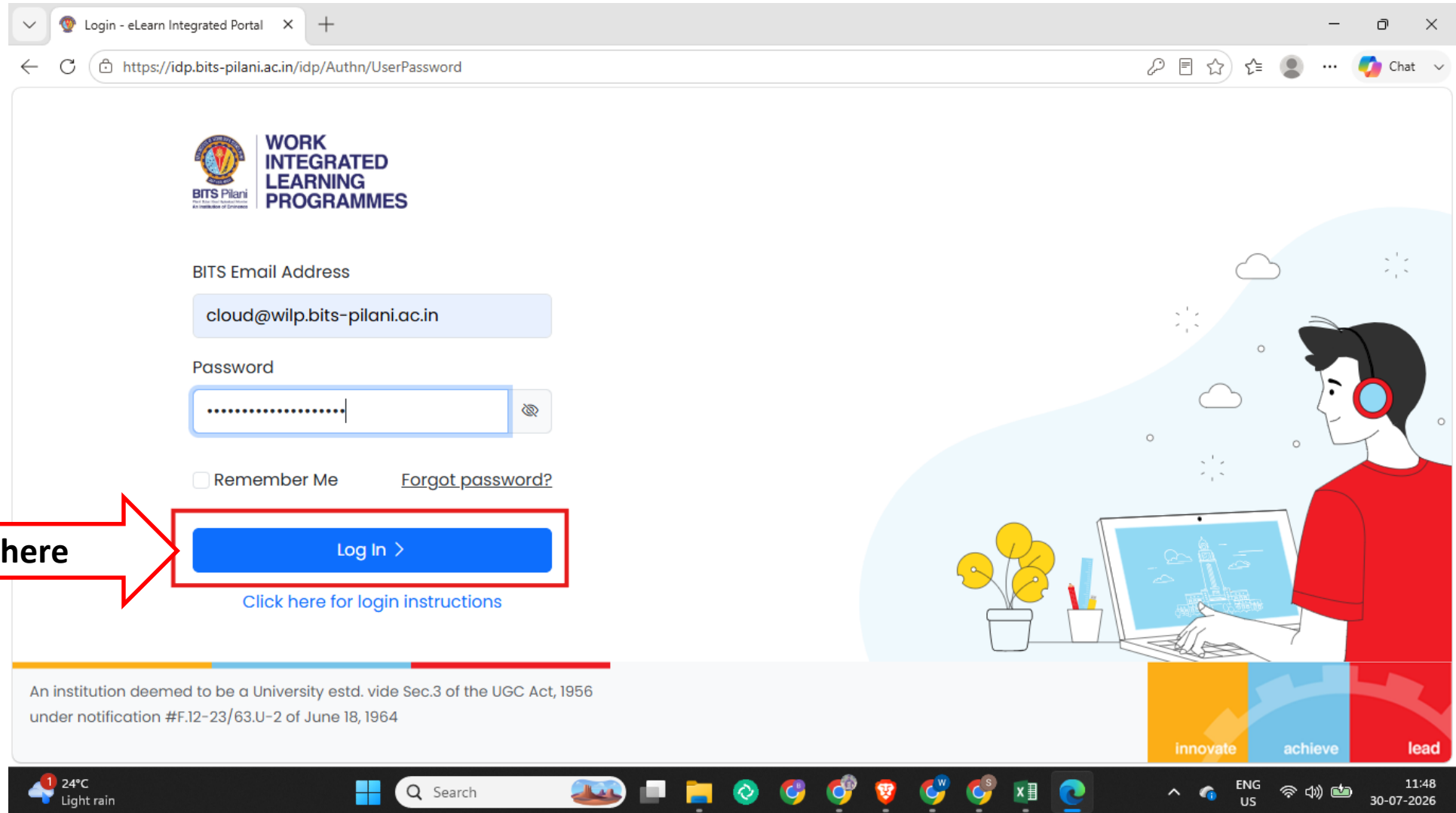
You'll be redirected to the BITS Pilani identity provider, then returned here.

Help Center • Privacy Policy • Contact IT

An institution deemed to be a University estd. vide Sec.3 of the UGC Act, 1956.

Step 2 - Login with BITS Credentials

Enter your BITS email ID and password of eLearn portal and click 'Log In'.



Login - eLearn Integrated Portal

https://idp.bits-pilani.ac.in/idp/Authn/UserPassword

 **WORK
INTEGRATED
LEARNING
PROGRAMMES**

BITS Email Address

cloud@wilp.bits-pilani.ac.in

Password

.....

☐ Remember Me [Forgot password?](#)

Click here → **Log In >**

[Click here for login instructions](#)

An institution deemed to be a University estd. vide Sec.3 of the UGC Act, 1956
under notification #F.12-23/63.U-2 of June 18, 1964

innovate achieve lead

24°C Light rain Search 11:48 30-07-2026

Step 3 - Launch Lab

Select your course and click 'Launch Lab'.

The screenshot displays the BITS Pilani Prayogshala Lab Management Portal dashboard. The interface includes a sidebar with the BITS Pilani logo and a 'Dashboard' button. The main content area features a grid of lab cards, each with a title, a list of technologies, and a 'Launch Lab' button. The cards are:

- PRE PROCESSING FOR AI READY DATA**: Technologies include NumPy, Pandas, SciPy, and +34 more.
- MULTIMODAL IR**: Technologies include Librosa, Open Cv, Langchain, and +3 more.
- ESSENTIAL MATHEMATICS FOR DATA SCIENCE**: Technologies include Octave and Matlab.
- MODERN DATABASES** (CSIST101): Technologies include Apache Flink, Apache Kafka, and Cassandra.
- SOFTWARE ENGINEERING FOR MACHINE LEARNING** (CSIST127): Technologies include Prefect, Prefect Cloud, and MIFlow.
- BIG DATA SYSTEMS** (CSIST135): Technologies include Hadoop, Spark, and Kafka.

Each card has a 'Launch Lab' button. A red box highlights the 'Cluster' button next to the 'Launch Lab' button for the 'BIG DATA SYSTEMS' card. A red arrow points to the 'Launch Lab' button for the 'SOFTWARE ENGINEERING FOR MACHINE LEARNING' card with the text 'Click here'. The bottom of the dashboard shows a 'Csis Student' profile and a database icon.

Step 4 - Launch Cluster Lab

Open your Cluster Lab session and click 'Launch all' to start all cluster nodes.

The screenshot shows a web browser window with the URL <https://stg.bitslabs.in/dashboard>. The page displays the 'Cluster Lab Session' interface for 'BIG DATA SYSTEMS - CSIST135'. A modal window is open, showing instructions under 'BEFORE YOU START' and a 'Launch all' button. A red arrow points to the 'Launch all' button with the text 'Click here'.

Cluster Lab Session
BIG DATA SYSTEMS - CSIST135

BEFORE YOU START

- Save to ~/workspace before stopping — everything outside it is permanently destroyed when the VM shuts down.
- 15-minute idle limit — your session auto-terminates after 15 min of inactivity. Any activity resets the clock.
- Keep this tab open — it sends your session heartbeat. Closing it may trigger early termination.
- 15-minute cluster timeout — if every node in the cluster is stopped, the cluster auto-terminates after 15 minutes.

Session: No Active Session

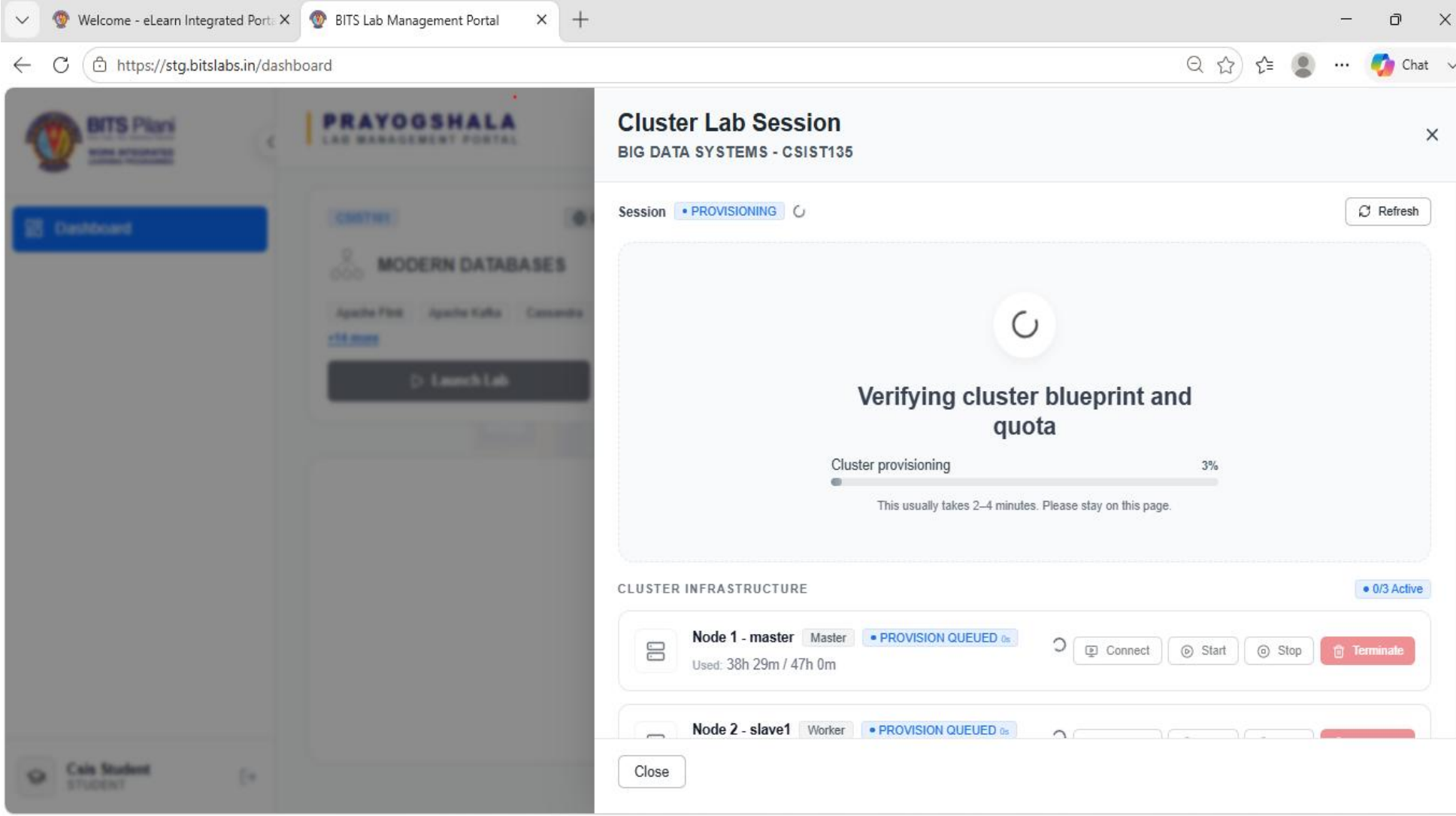
CLUSTER INFRASTRUCTURE 0/3 Active

Node	Role	Used	Actions
Node 1 - master	Master	Used: 38h 29m / 47h 0m	Connect Start Stop Terminate
Node 2 - slave1	Worker	Used: 42h 7m / 47h 0m	Connect Start Stop Terminate
Node 3 - slave2	Worker	Used: 42h 20m / 47h 0m	Connect Start Stop Terminate

Close

Step 5 - Cluster Provisioning

After launching the cluster, the portal displays the provisioning status. Wait while the cluster nodes are provisioned.



The screenshot shows a web browser window with the URL <https://stg.bitslabs.in/dashboard>. The page displays the "Cluster Lab Session" for "BIG DATA SYSTEMS - CSIST135". The session status is "PROVISIONING". A progress bar indicates "Cluster provisioning" is at 3%. Below the progress bar, it states "This usually takes 2-4 minutes. Please stay on this page." The "CLUSTER INFRASTRUCTURE" section shows two nodes: "Node 1 - master" (Master) and "Node 2 - slave1" (Worker). Both nodes are in the "PROVISION QUEUED" state. The "Node 1 - master" node shows "Used: 38h 29m / 47h 0m". The "Node 2 - slave1" node shows "Used: 0h 0m / 47h 0m". The "Node 1 - master" node has buttons for "Connect", "Start", "Stop", and "Terminate". The "Node 2 - slave1" node has buttons for "Connect", "Start", "Stop", and "Terminate". A "Close" button is located at the bottom left of the cluster infrastructure section.

Cluster Lab Session
BIG DATA SYSTEMS - CSIST135

Session **PROVISIONING**

Refresh

Verifying cluster blueprint and quota

Cluster provisioning 3%

This usually takes 2-4 minutes. Please stay on this page.

CLUSTER INFRASTRUCTURE 0/3 Active

Node	Role	Status	Used	Buttons
Node 1 - master	Master	PROVISION QUEUED 0s	Used: 38h 29m / 47h 0m	Connect, Start, Stop, Terminate
Node 2 - slave1	Worker	PROVISION QUEUED 0s	Used: 0h 0m / 47h 0m	Connect, Start, Stop, Terminate

Close

Step 6 - Refresh Cluster Status

If the provisioning status does not update automatically after provisioning bar disappears, click 'Refresh' to retrieve the latest cluster status.

The screenshot displays the BITS Lab Management Portal interface. The main panel shows the 'Cluster Lab Session' for 'BIG DATA SYSTEMS - CSIST135'. The session status is 'PROVISIONING'. A yellow box titled 'BEFORE YOU START' contains instructions: 'Save to ~/workspace before stopping', '15-minute idle limit', 'Keep this tab open', and '15-minute cluster timeout'. A 'Refresh' button is highlighted with a red box. Below this, the 'CLUSTER INFRASTRUCTURE' section shows three nodes: 'Node 1 - master' (Master), 'Node 2 - slave1' (Worker), and 'Node 3 - slave2' (Worker). All nodes are in a 'PROVISION QUEUED' state. Each node has a 'Connect', 'Start', 'Stop', and 'Terminate' button. The 'Used' time for each node is displayed as 38h 29m / 47h 0m for the master and 42h 7m / 47h 0m for the slaves. A 'Close' button is visible at the bottom left of the modal.

Step 7 - Cluster Running

When the cluster is running, the portal displays the shared cluster credentials and the status of each node.

The screenshot shows the BITS Lab Management Portal interface. The main window displays the 'Cluster Lab Session' for 'BIG DATA SYSTEMS - CSIST135'. The session status is 'RUNNING'. A yellow warning banner indicates a '15-minute cluster timeout — if every node in the cluster is stopped, the cluster auto-terminates after 15 minutes.' Below this, there are buttons for 'Refresh', 'Start all', 'Stop all', and 'Terminate all'. The 'SHARED CLUSTER CREDENTIALS' section shows the 'Username' as 'csis_student' and the 'Password' as a masked field. The 'CLUSTER INFRASTRUCTURE' section shows three active nodes: 'Node 1 - master' (Master, RUNNING, 17m 6s), 'Node 2 - slave1' (Worker, RUNNING, 17m 6s), and 'Node 3 - slave2' (Worker, RUNNING, 17m 6s). Each node has a 'Connect' button and 'Start', 'Stop', and 'Terminate' buttons. The background shows the main dashboard with 'Welcome back, Csis Student' and 'My Laboratories' section.

Cluster Lab Session
BIG DATA SYSTEMS - CSIST135

15-minute cluster timeout — if every node in the cluster is stopped, the cluster auto-terminates after 15 minutes.

Session: **RUNNING** [Refresh] [Start all] [Stop all] [Terminate all]

SHARED CLUSTER CREDENTIALS

Username: csis_student Password: [Masked]

CLUSTER INFRASTRUCTURE 3/3 Active

Node	Role	Status	Time	Actions
Node 1 - master	Master	RUNNING	17m 6s	[Connect] [Start] [Stop] [Terminate]
Node 2 - slave1	Worker	RUNNING	17m 6s	[Connect] [Start] [Stop] [Terminate]
Node 3 - slave2	Worker	RUNNING	17m 6s	[Connect] [Start] [Stop] [Terminate]

Close

Step 8 – Connecting to the VMs

Click on Connect button to access the individual Nodes upon your choice of work

The screenshot shows the BITS Lab Management Portal interface. The main panel displays the 'Cluster Lab Session' for 'BIG DATA SYSTEMS - CSIST135'. The session status is 'RUNNING'. A yellow warning banner indicates a 15-minute cluster timeout. Below this, there are controls for the session: 'Refresh', 'Start all', 'Stop all', and 'Terminate all'. The 'SHARED CLUSTER CREDENTIALS' section shows the username 'csis_student' and a masked password. The 'CLUSTER INFRASTRUCTURE' section shows three nodes: 'Node 1' (IP: 10.2.11.87), 'Node 2 - slave1' (IP: 10.2.11.87, Worker, RUNNING 2s), and 'Node 3 - slave2' (IP: 10.2.11.145, Worker, RUNNING 2s). Each node has a 'Connect' button, a 'Start' button, a 'Stop' button, and a 'Terminate' button. A red arrow points to the 'Connect' button for Node 1, with the text 'Click here' next to it.

Cluster Lab Session
BIG DATA SYSTEMS - CSIST135

15-minute cluster timeout — if every node in the cluster is stopped, the cluster auto-terminates after 15 minutes.

Session **RUNNING** Refresh Start all Stop all Terminate all

SHARED CLUSTER CREDENTIALS

Username Password

csis_student

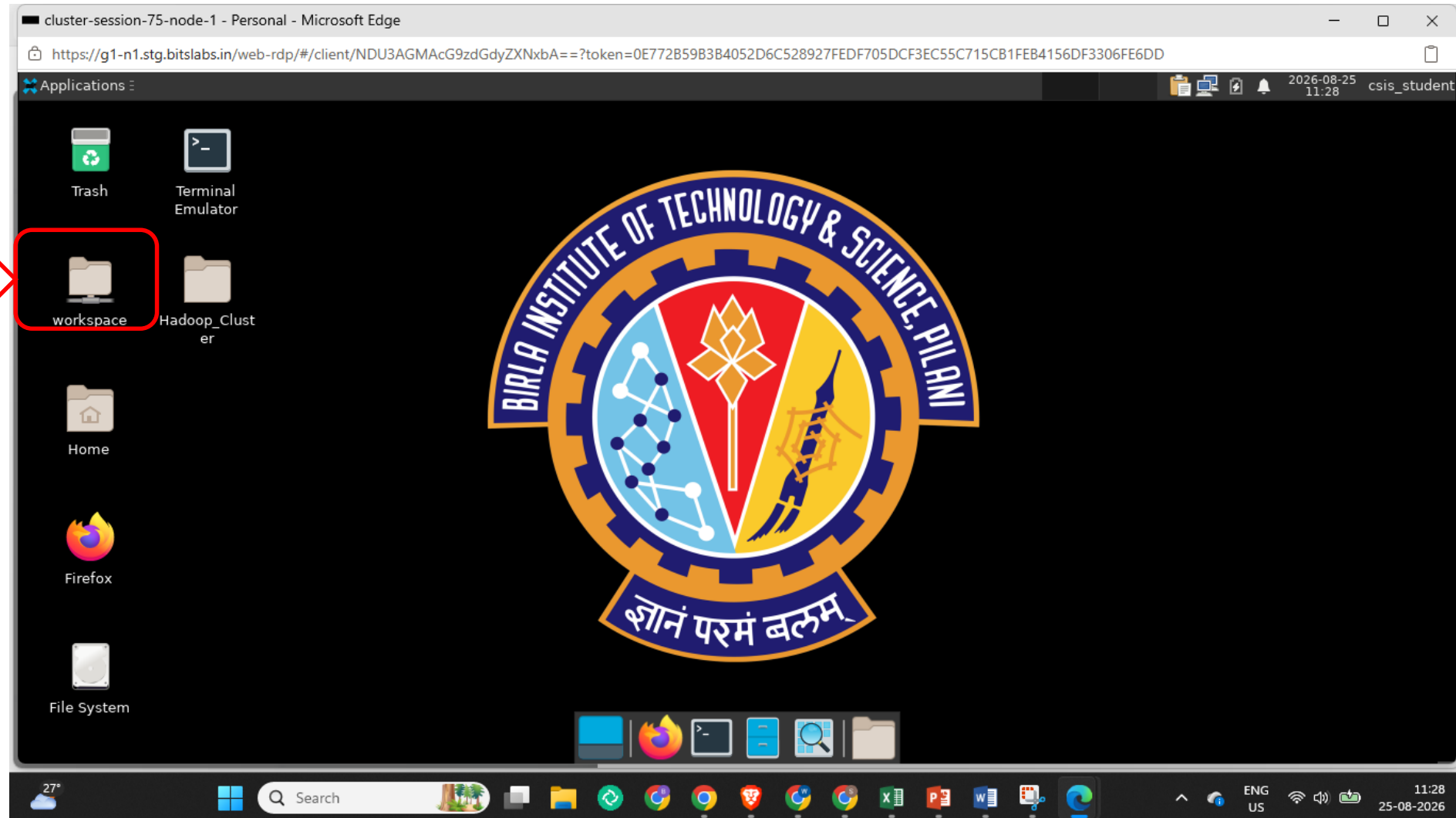
CLUSTER INFRASTRUCTURE **3/3 Active**

Node	IP	Role	Status	Connect	Start	Stop	Terminate
Node 1	10.2.11.87			Click here			
Node 2 - slave1	10.2.11.87	Worker	RUNNING 2s				
Node 3 - slave2	10.2.11.145	Worker	RUNNING 2s				

Close

Step 9- Virtual Desktop upon clicking on Connect button(This is only one Node just for reference)

Work on the virtual desktop. Save files in the workspace folder(Persistent folder) to backup your files for further use. The files saved in workspace folder can be accessed across all the nodes in the cluster.



Step 10 - Focus on a Cluster Node

Click 'Focus' for the required node to work with or access that specific cluster node.

The screenshot shows the BITS Lab Management Portal interface. The main panel displays the 'Cluster Lab Session' for 'BIG DATA SYSTEMS - CSIST135'. The session status is 'RUNNING'. Below this, there are fields for 'Username' (csis_student) and 'Password' (masked). A 'Connection token expires at 11/8/2026, 4:40:51 pm' is also shown. The 'CLUSTER INFRASTRUCTURE' section lists three nodes: 'Node 1', 'Node 2 - slave1', and 'Node 3 - slave2'. Each node has a 'Focus' button, a 'Start' button, a 'Stop' button, and a 'Terminate' button. A red arrow points to the 'Focus' button for 'Node 2 - slave1'.

Cluster Lab Session
BIG DATA SYSTEMS - CSIST135

Session **RUNNING** [Refresh] [Start all] [Stop all] [Terminate all]

SHARED CLUSTER CREDENTIALS

Username: csis_student Password: [Masked]

Connection token expires at 11/8/2026, 4:40:51 pm

CLUSTER INFRASTRUCTURE **3/3 Active**

Node	Role	Status	IP	Used	Focus	Start	Stop	Terminate
Node 1	Worker	RUNNING	10.2.11.123	2h 0m / 22h 0m	Click here	[Start]	[Stop]	[Terminate]
Node 2 - slave1	Worker	RUNNING	10.2.11.123	2h 0m / 22h 0m	[Focus]	[Start]	[Stop]	[Terminate]
Node 3 - slave2	Worker	RUNNING	10.2.11.15	2h 1m / 22h 0m	[Focus]	[Start]	[Stop]	[Terminate]

Close

Step 11 - Stop All Cluster Nodes(cluster state will be retained for 15mins only)

- 1.This will stop all running nodes but the state of the cluster machines will be preserved. If all machines are stopped, the lab would be auto terminated after 15min.
2. You can also stop individual machines as well if you need only few machines in the cluster to work with and this will not terminate the cluster.

The screenshot shows the BITS Lab Management Portal interface. The main panel displays the 'Cluster Lab Session' for 'BIG DATA SYSTEMS - CSIST135'. The session status is 'RUNNING'. A red arrow points to the 'Stop all' button, with the text 'Click here' next to it. Below the session status, there are fields for 'Username' (csis_student) and 'Password' (masked). A 'Connection token expires at 11/8/2026, 4:40:51 pm' is also shown. The 'CLUSTER INFRASTRUCTURE' section shows three nodes: 'Node 1 - master' (Master, RUNNING), 'Node 2 - slave1' (Worker, RUNNING), and 'Node 3 - slave2' (Worker, RUNNING). Each node has a 'Focus', 'Start', 'Stop', and 'Terminate' button. The 'Stop' buttons for all three nodes are highlighted with red boxes.

Node	Role	Status	IP	Used	Buttons
Node 1 - master	Master	RUNNING	10.2.11.163	1h 46m / 22h 0m	Focus, Start, Stop, Terminate
Node 2 - slave1	Worker	RUNNING	10.2.11.123	2h 0m / 22h 0m	Focus, Start, Stop, Terminate
Node 3 - slave2	Worker	RUNNING	10.2.11.15	2h 1m / 22h 0m	Focus, Start, Stop, Terminate

Step 12 - Terminate All Cluster Nodes

Click 'Terminate all' only after completing the lab. Termination permanently releases the cluster nodes. Only the data inside ~/workspace directory will be preserved.

The screenshot shows the BITS Lab Management Portal interface. The main panel displays the 'Cluster Lab Session' for 'BIG DATA SYSTEMS - CSIST135'. The session status is 'RUNNING'. Below this, there are fields for 'Username' (csis_student) and 'Password' (masked). A 'Terminate all' button is highlighted with a red box and a red arrow pointing to it, with the text 'Click here' next to the arrow. Below the password field, it says 'Connection token expires at 11/8/2026, 4:40:51 pm'. The 'CLUSTER INFRASTRUCTURE' section shows three nodes: 'Node 1 - master' (Master, RUNNING), 'Node 2 - slave1' (Worker, RUNNING), and 'Node 3 - slave2' (Worker, RUNNING). Each node has a 'Focus', 'Start', 'Stop', and 'Terminate' button. The 'Terminate' button for each node is highlighted with a red box. A 'Close' button is at the bottom left of the panel.

Cluster Lab Session
BIG DATA SYSTEMS - CSIST135

Session • RUNNING

SHARED CLUSTER CREDENTIALS

Username Password

csis_student

Connection token expires at 11/8/2026, 4:40:51 pm

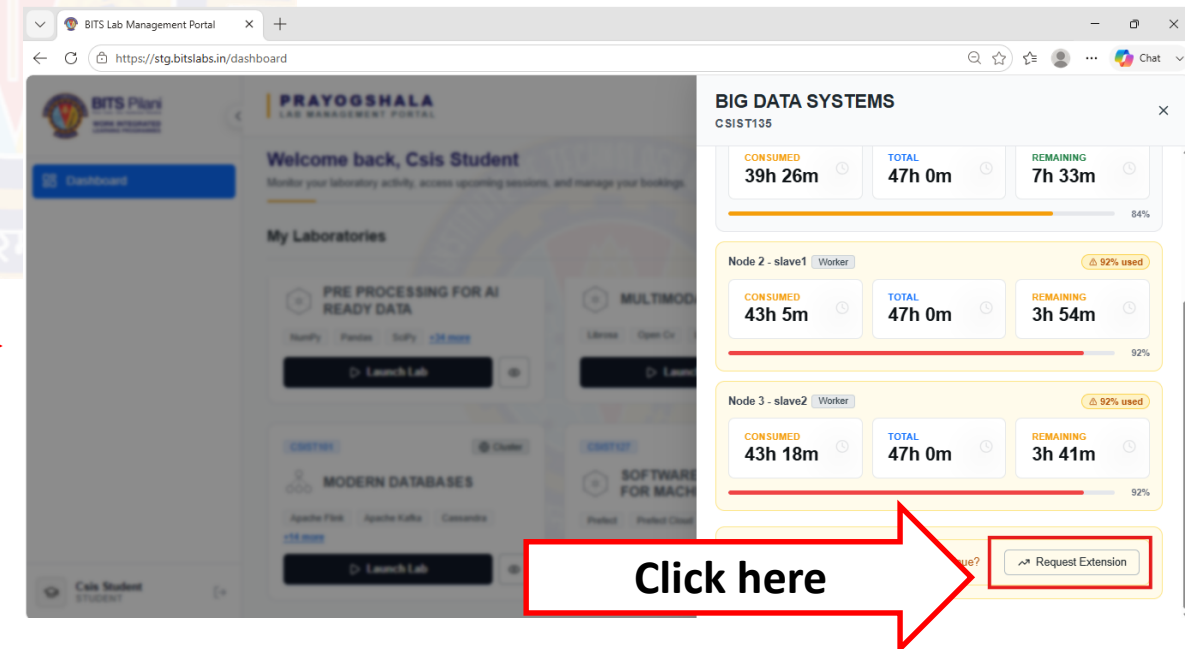
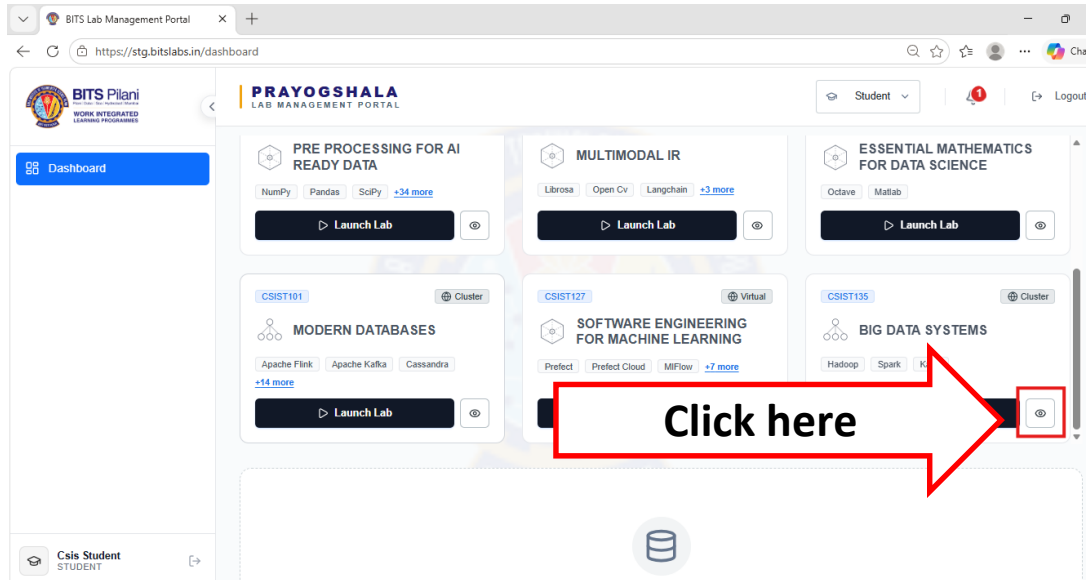
CLUSTER INFRASTRUCTURE • 3/3 Active

Node	Role	Status	IP	Used	Actions
Node 1 - master	Master	RUNNING	10.2.11.163	1h 46m / 22h 0m	Focus Start Stop Terminate
Node 2 - slave1	Worker	RUNNING	10.2.11.123	2h 0m / 22h 0m	Focus Start Stop Terminate
Node 3 - slave2	Worker	RUNNING	10.2.11.15	2h 1m / 22h 0m	Focus Start Stop Terminate

Close

Step 13 – Quota Extension Request

The option will be enabled only when you have used 85% of the allocated duration.



Important instructions

-  Save your work in the /workspace folder before stopping or terminating the cluster.
-  Use 'Refresh' if the cluster status is not updated on the screen.
-  Use 'Focus' to work with the required cluster node.
-  Use 'Stop all' to stop the cluster when you want to pause and continue later before 15mins. If all the machines are stopped, the cluster will be auto terminated. You can also use STOP button to stop individual machines if you need only few machines in the cluster.
-  Use 'Terminate all' only when you have completed the lab and no longer need the cluster.

Technical Support details

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Thank You